

A08 Retail Industry Module Reflections

Industry Overview

The supply chain industry is all about getting products from the company that makes them to the customer. This includes ordering materials, making products, storing them in warehouses, and delivering them to stores or people's homes. Almost every business depends on a supply chain, like Amazon, Walmart, hospitals, grocery stores, and even car companies.

This industry is really important because people expect to get what they order quickly. If products are delayed or out of stock, companies lose money and customers get frustrated. That's why more businesses are using AI to help make better decisions and keep everything running smoothly.

Current AI Trends

One way companies use AI is for demand forecasting. This means AI looks at old sales, shopping trends, holidays, and even the weather to predict what people will want to buy. Companies like Amazon and Walmart use this so they can stock up on the right products before customers order them. This helps keep shelves full and cuts down on waste.

Another way AI is used is inventory optimization. Instead of guessing how much inventory they need, AI helps companies figure out the right amount to keep in stock. This saves money because businesses don't have too much or too little inventory. It also helps customers get their orders faster.

AI is also helping with delivery routes. Companies like UPS use AI to find the fastest routes for drivers by looking at traffic, weather, and road conditions. This helps save gas, deliver packages faster, and lower costs.

AI is also being used to keep an eye on suppliers and predict problems before they happen. If there's bad weather, a strike, or another issue that could slow down deliveries, AI can warn companies early. That gives them time to find another supplier or come up with a backup plan.

Ethical Implications

Even though AI has a lot of benefits, it also comes with some problems. One issue is that some jobs could disappear. As warehouses and factories use more robots and automation, fewer workers may be needed for certain tasks. On the other hand, new jobs will be created for people who know how to work with AI, but not everyone has those skills yet.

Another concern is privacy. AI needs a lot of data to work well. Companies collect information about customers, employees, and suppliers. If that information isn't protected, it could be stolen or used the wrong way. Businesses have to make sure they keep people's information safe.

Another issue is that AI doesn't always explain how it makes decisions. For example, it might recommend using a different supplier or ordering less inventory without saying why. That can make it hard for managers to know if the decision is actually the right one. I think people should still be involved when making important decisions instead of letting AI do everything.

Future Trends

I think AI is only going to become more common in supply chains over the next five to ten years. Companies will get even better at predicting what customers want, which should help reduce shortages and wasted products.

We'll probably also see more robots working in warehouses and maybe even self-driving delivery vehicles in some places. AI will help companies respond faster when something unexpected happens, like a storm or a supply shortage.

At the same time, companies will have to be careful about cybersecurity and protecting customer information. Workers will also need training so they can keep up with all the new technology. Businesses that use AI the right way while still relying on people for important decisions will probably do the best.

My Reflection

The biggest thing I learned from this assignment is that AI does a lot more than I thought. Before this, I mostly thought AI was used for things like chatbots or answering questions online. I didn't realize it also helps companies predict sales, manage inventory, plan delivery routes, and avoid supply chain problems before they happen.

I think I would be interested in working in this industry because technology is becoming a bigger part of business every year. It seems like there will be a lot of opportunities for people who understand both business and AI. At the same time, I don't think AI should replace people completely. There are still situations where human judgment is important. Overall, I think AI can make supply chains faster and more efficient as long as companies use it responsibly.